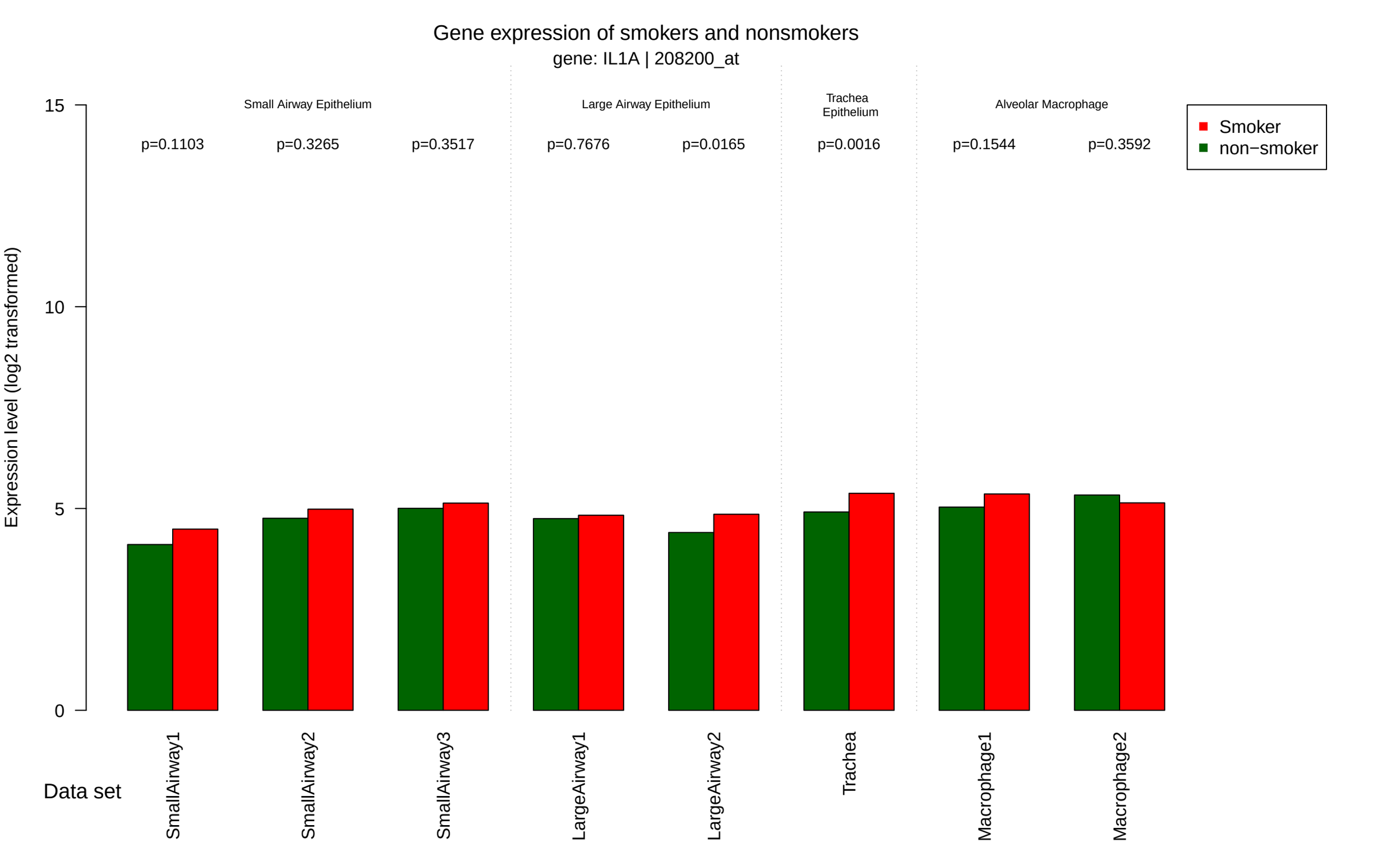
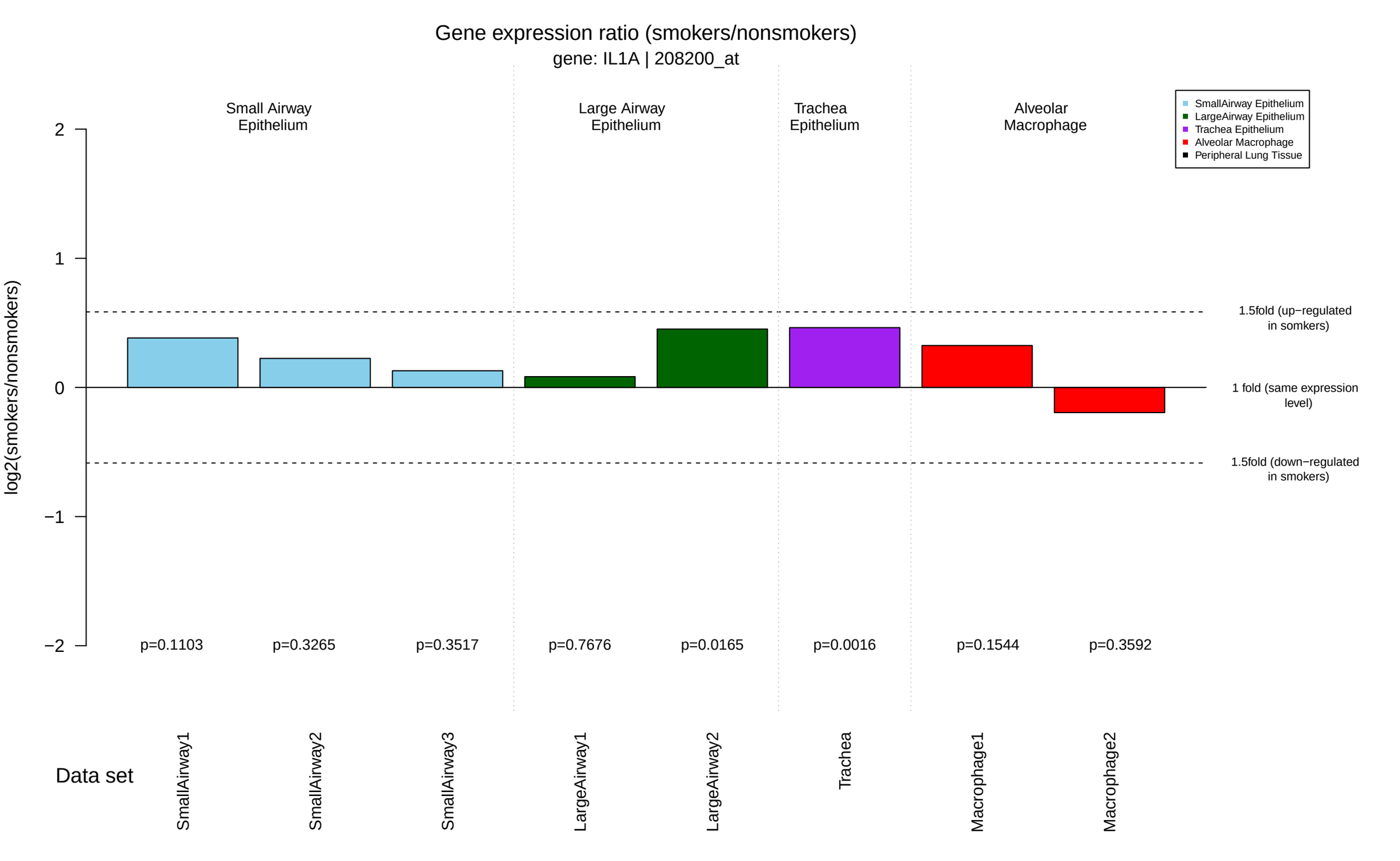
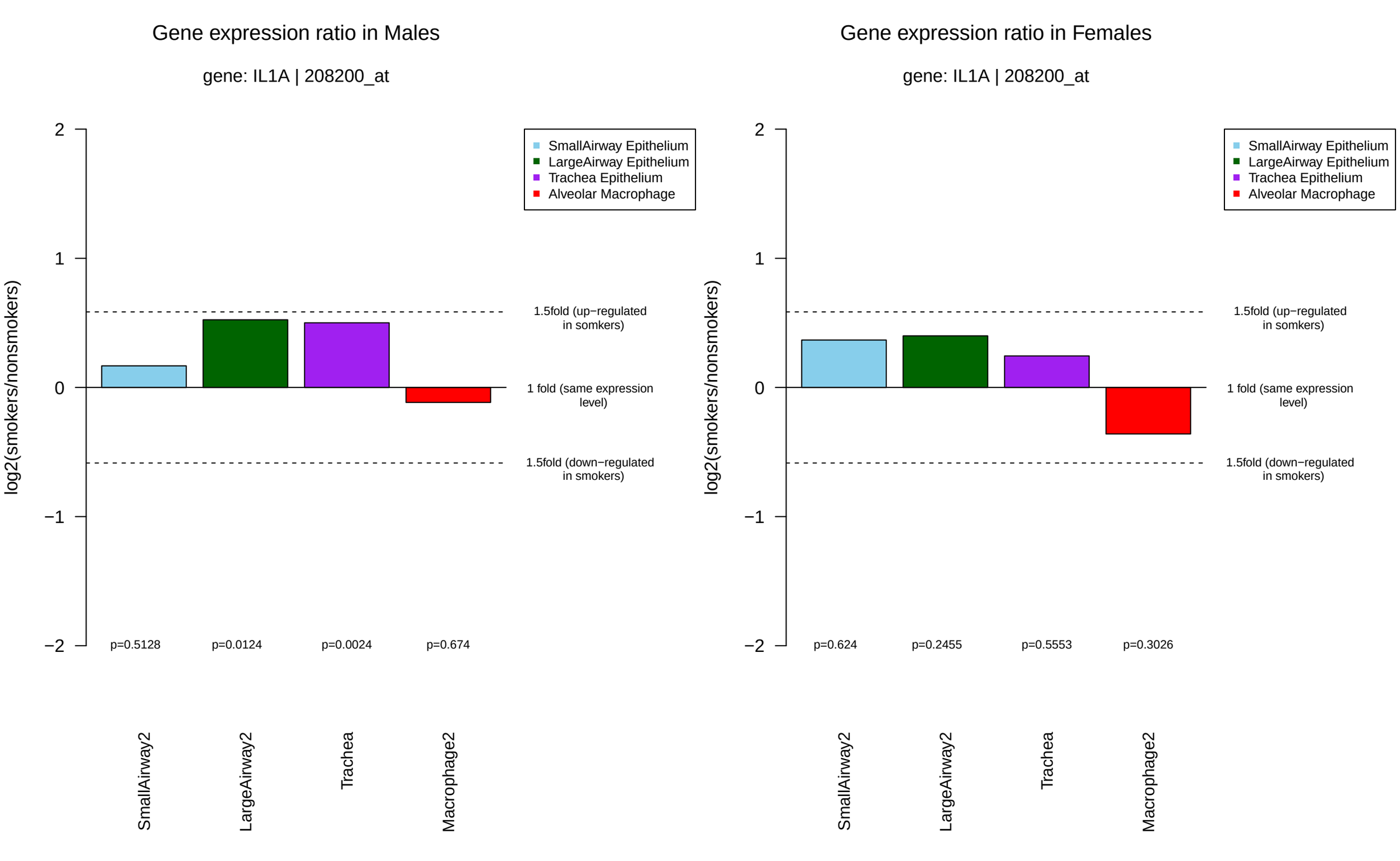
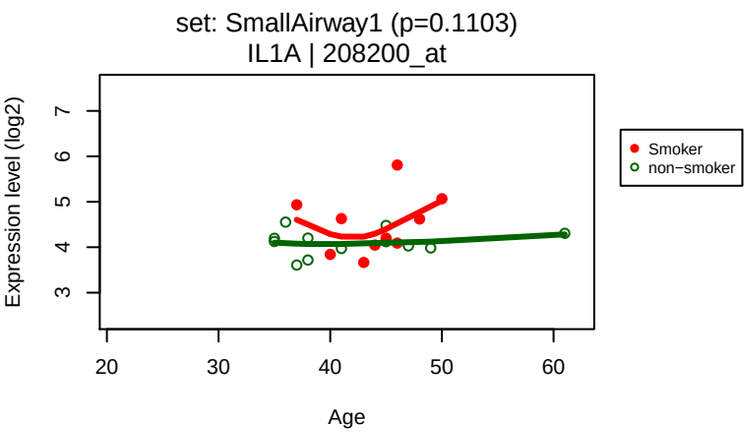
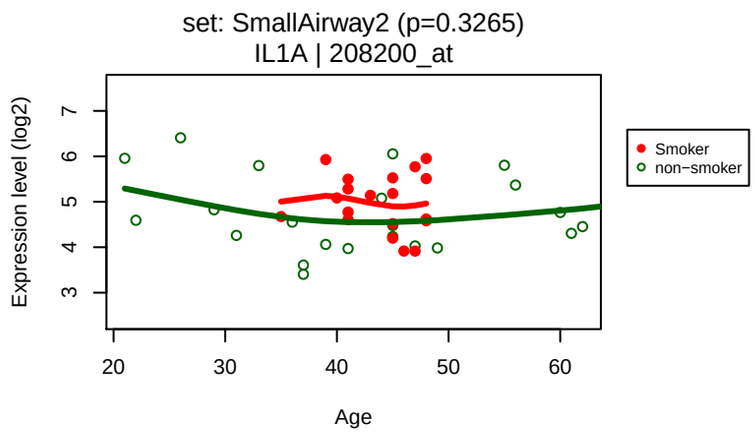
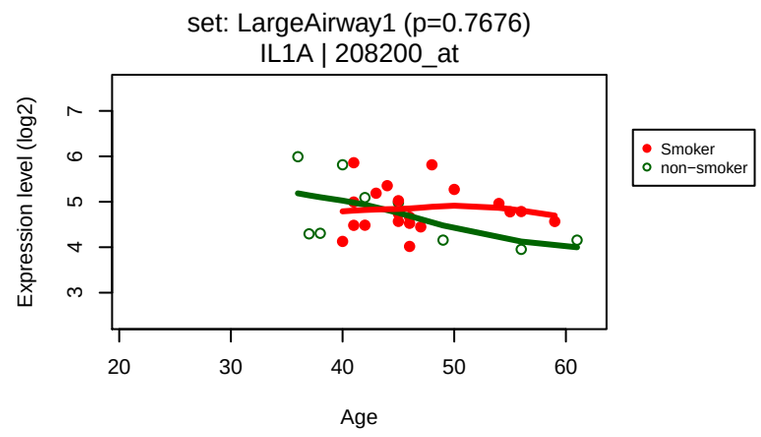

Gene expression of smokers and nonsmokers
gene: IL1A | 208200_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1103
p=0.3265
p=0.3517
p=0.7676
p=0.0165
p=0.0016
p=0.1544
p=0.3592
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

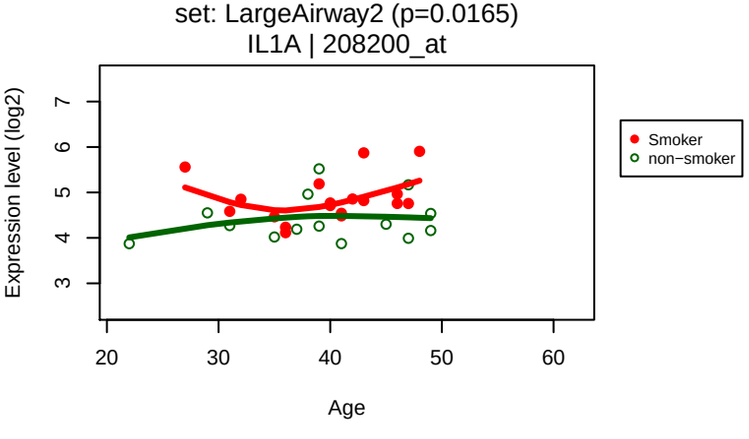

Gene expression ratio (smokers/nonsmokers)
gene: IL1A | 208200_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1103
p=0.3265
p=0.3517
p=0.7676
p=0.0165
p=0.0016
p=0.1544
p=0.3592
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

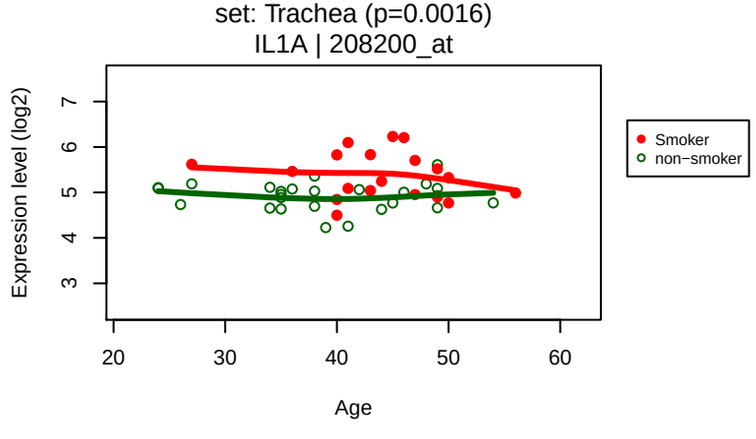

Gene expression ratio in Males
gene: IL1A | 208200_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5128
p=0.0124
p=0.0024
p=0.674
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL1A | 208200_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.624
p=0.2455
p=0.5553
p=0.3026
SmallAirway2
LargeAirway2
Trachea
Macrophage2

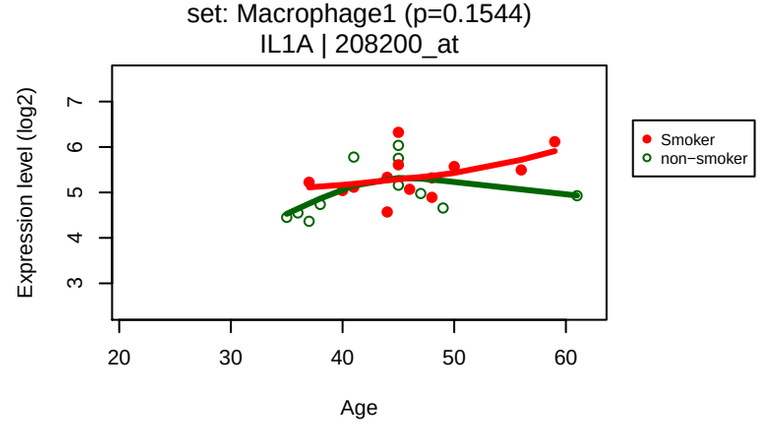

set: SmallAirway1 (p=0.1103)
IL1A | 208200_at
Expression level (log2)
Age
Smoker
non−smoker

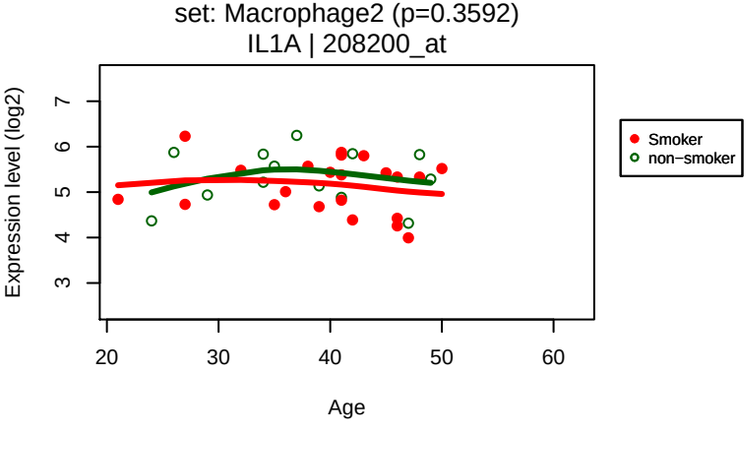

set: SmallAirway2 (p=0.3265)
IL1A | 208200_at
Expression level (log2)
Age
Smoker
non−smoker

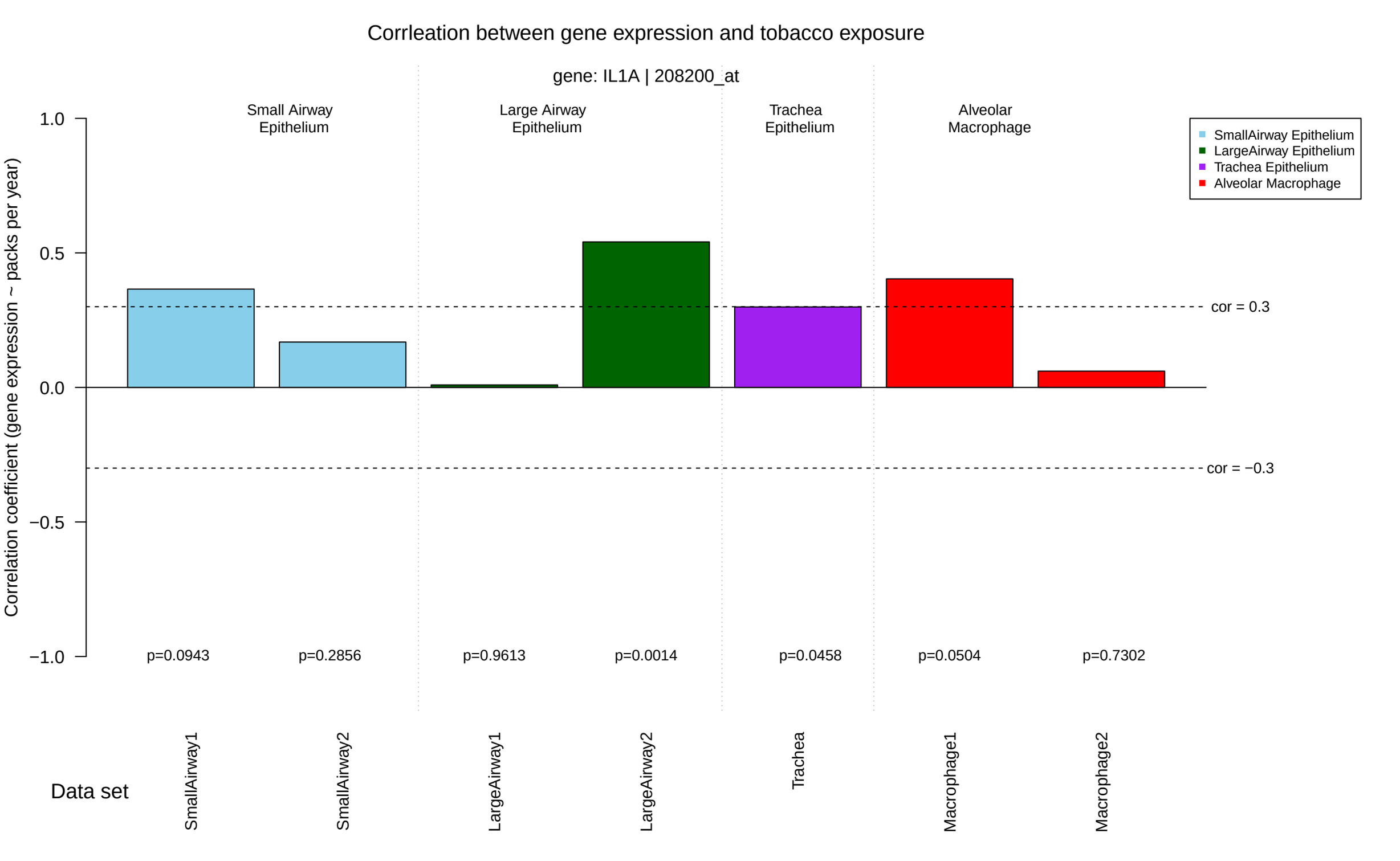

set: LargeAirway1 (p=0.7676)
IL1A | 208200_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0165)
IL1A | 208200_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0016)
IL1A | 208200_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1544)
IL1A | 208200_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3592)
IL1A | 208200_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL1A | 208200_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0943
p=0.2856
p=0.9613
p=0.0014
p=0.0458
p=0.0504
p=0.7302
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2